

ARTICLE

Crop Economics, Production, and Management

Cover crop mixtures enhance stability but not productivity in a semi-arid climate

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Abstract

Cover crops are a critical component of sustainable agroecosystems, yet their performance in semi-arid environments with short growing seasons and highly variable precipitation is inconsistent. To evaluate productivity and stability of single species vs. multi-species cover crop plantings with variable water input, cover crops were evaluated at two sites in North Dakota under dryland and dryland plus irrigation conditions. Cover crop treatments consisted of three warm-season monocultures, a three warm-season species mixture (MIX3), and a three warm-season and three cool-season species mixture (MIX6). Averaged across year and water regime, biomass production was not greater for mixtures (8,693 kg ha⁻¹, MIX3; 10,212 kg ha⁻¹, MIX6) than for the most productive monoculture (11,295 kg ha⁻¹, sunflower [*Helianthus annuus* L.]) at the West site. The same held true at the East site (7,486 kg ha⁻¹, MIX3; 8,603 kg ha⁻¹, MIX6), with cowpea [*Vigna unguiculata* (L.) Walp.] producing the most biomass, 10,886 kg ha⁻¹, of all monocultures. At the West site, variability in production was lowest for MIX6 when pooled across year and water regime (CV = 25%) compared to monocultures (CV ≥ 49%). At the East site, cowpea had the lowest CV (24%) while mixtures averaged 35%. Pooled across sites, production in the MIX6 plots had the lowest variability (29%). Grasses consistently contributed to high proportions of total biomass production compared to other functional types. Multi-species cover crop mixtures are more versatile than single species plantings across climates, and resilient cover crop mixtures integrated into semi-arid crop rotations as full- season plantings should include warm and cool season grasses.

1 | INTRODUCTION

The benefits of cover crops and their role in creating multi-functional agricultural landscapes that support a wide range of agronomic and ecological services has been well documented (Baraibar et al., 2018; Biederbeck et al., 1996; Blanco-Canqui et al., 2013; Dabney et al., 2001; Finney & Kaye, 2017; Finney

et al., 2017; Fisk et al., 2001; Liebig et al., 2015; Liu et al., 2005; Mallinger et al., 2019; Mazzoncini et al., 2011; Norsworthy et al., 2011; Schipanski et al., 2014; Snapp et al., 2005; Teasdale, 1996; Tonitto et al., 2006; Wiggins et al., 2015). According to the most recent national survey of cover crop users, the average number of cover crop land area planted per farm increased from 136 to 188 ha (337 to 465 acres) (a 40% increase) between 2015 and 2019 (CTIC & SARE, 2020). Of the producers responding to the survey, those that received payments from federal, state, local soil and water

Abbreviations: CV, coefficient of variation; MIX3, 3-species cover crop mixture; MIX6, 6-species cover crop mixtures; NGP, northern Great Plains; VWC, volumetric water content; WUE, water use efficiency.

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conservation district, or other entities to plant cover crops increased from 37% in 2014 to 52% in 2019. Whether due to greater awareness of cover crop benefits, greater availability of incentive programs, or other factors such as low commodity prices, these statistics indicate growing interest in and adoption of cover crops across U.S. farms (CTIC & SARE, 2020).

Farmers are encouraged to plant multi-species cover crop mixtures instead of single species plantings to enhance their tolerance to adverse environmental conditions (Clark, 2007; Jay Fuhner [NRCS-Retired], personal communication, 2016). In natural, perennial ecosystems, diverse plant communities reduce long-term interannual biomass production variability, including during periods of drought (Tilman, 1996). When applied to annual cover crop plantings, some researchers have suggested that complex cover crop mixtures are either more productive (Khan & McVay, 2019; Sanderson et al., 2018) or can increase the number of services provided as compared to single species cover crops (Finney & Kaye, 2017; Finney et al., 2016; Finney et al., 2017; Schipanski et al., 2014), and can do so without adversely impacting cash crop yields (Hunter et al., 2019; Nielsen et al., 2016). Still others have found mixed results, with cover crop mixture productivity and related ecological services such as forage production, weed suppression, and soil water retention dependent on several interrelated factors such as cover crop mixture component species, cover crop termination methods, relative plantings dates, subsequent cash crop species, and other factors (Baraibar et al., 2018; Sanderson et al., 2018; Smith et al., 2014; Wortman et al., 2012).

In the northern Great Plains (NGP) region of the United States, limited or erratic precipitation as well as a colder climate with a shorter growing season pose barriers to cover crop adoption due to the economic risks (either in terms of lack of productivity from cover crops or through yield reductions to the subsequent cash crop) associated with their adoption in semi-arid climates (CTIC & SARE, 2017; Liebig et al., 2015; Mitchell et al., 2015; Sanderson et al., 2018; Unger & Vigil, 1998). In the NGP, greater production and less variability have been reported when using multi-species cover crop mixtures when compared with single species cover crop plantings (Khan & McVay, 2019). These findings are supported by cover crop research conducted at Mandan, ND, which demonstrated that multi-species mixtures productivity that included plant functional type redundancy were less variable than single species plantings under variable precipitation (Franco, unpublished data, 2019). Sanderson et al. (2018) also reported 30–60% more forage for multi-species mixtures than single species plantings, but not greater than the most productive spring seeded monoculture cover crops. However, when seeded in the fall, erratic and limited rainfall coupled with the short growing season in the NGP resulted in no differences among mixtures and monocultures based on weed suppression and forage production. Liebig et al. (2015)

Core Ideas

- Cover crop performance is limited by inconsistent precipitation in semi-arid climates.
- Cover crop mixtures did not out-yield the most productive single species plantings.
- A six-species cover crop mixture resulted in more stable cover crop yields across sites and years.
- Grasses contributed greater biomass and more consistently to mixture production.
- Kale productivity increased in the dry year making it a good choice for drier climates.

reported up to 93% decline in cover crop production and associated reduction in short-term soil responses in years when establishment was hindered by timely precipitation in the NGP. They attributed cover crop productivity inconsistency to variability in weather conditions and not to differences between multi-species and single species plantings. In the central Great Plains, Nielsen et al. (2015a) and Nielsen et al. (2016) reported that the added expense of planting cover crop mixtures was justified when a certain forage quality is desired, which can be achieved with mixtures, but was not justified based on measured responses including cash crop yield, residue cover, and aboveground biomass production.

With cover crop mixtures increasing in popularity (more than 76% of producers indicated they either planted mixtures in 2019, intended to plant mixtures in 2020, or planted mixtures in 2019 and plan to do so again in 2020 [CTIC & SARE, 2020]), and timely precipitation is a major concern in the semi-arid NGP, evaluation of cover crop performance under variable climates is critical for their successful integration into annual cropping systems. The few studies that have compared multi-species mixtures with monocultures under varied water input in the semi-arid Great Plains showed similar productivity between the two (Nielsen et al., 2015a; Nielsen et al., 2015b; Franco, unpublished data, 2019), but less variability associated with mixtures (Franco, unpublished data, 2019). We have found only two studies have compared multi-species and single species productivity under dryland and irrigated conditions to simulate variable water input and climates in a semi-arid environment. Only Nielsen et al. (2015a) and Nielsen et al. (2015b) compared these variables at more than one site in the lower latitudes of the Central Great Plains (USDA Hardiness Zones 5a–5b; Köppen–Geiger climate classification Bsk, cold semi-arid climate). However, no studies were reported in the more northern climate of North Dakota (USDA Hardiness Zone 4a; Köppen–Geiger Climate Classification Dfb, humid continental mild summers) with a shorter growing season. Thus, the overall objective

of this study was to evaluate productivity of single species cover crop plantings and cover crop mixtures under variable rainfall and irrigation inputs at two sites in the semi-arid NGP (North Dakota). Because it is challenging to establish a cover crop late in the season after a cash crop in this area, growers often plant full-season cover crops in rotation with cash crops (Carr et al., 2012), as in this study. Further, we chose to focus on productivity because overall biomass correlates with several critical cover crop services, including weed suppression, soil nutrient management (via nutrient production or uptake/retention), soil erosion mitigation, and soil organic matter provision (Finney et al., 2016; Florence et al., 2019), though we recognize that sufficient biomass production must be generated to provide these services.

We hypothesized that multi-species mixtures would be more productive and less variable across sites and years than single species plantings, and that these differences would be magnified under drier conditions. Additionally, soil macronutrients were evaluated at one site to assess N, P, and K relative differences among single species and multi-species cover crop plantings at the end of the growing season.

2 | MATERIALS AND METHODS

2.1 | Site descriptions

This study was conducted near Mandan, ND (46.81° N, −100.92° W) at the USDA-Agriculture Research Service Northern Great Plains Research Laboratory (NGPRL), which served as the western site (West), and near Absaraka and Fargo, ND (46.99° N, −97.35° W and 46.90° N, −96.82° W, respectively) at the North Dakota State University (NDSU) Horticultural Research Farm near Absaraka, ND, in 2016 and NDSU campus research plots in Fargo, ND, during 2017, which collectively served as the eastern site (East). Due to irrigation inadequacy, the East site was moved to the Fargo location, which is 64 km from the Absaraka location. The West and East sites are classified as two distinct agro-ecoregions (Padbury et al., 2002).

The West site has a long history of no-till management with conventional fertilizer and pesticide inputs. The West site has a long-term (1981–2010) mean annual precipitation of 401 mm and mean annual temperature of 5.2 °C (Padbury et al., 2002). The dominant soil type is a Temvik–Wilton silt loam (fine-silty, mixed, superactive, frigid Typic and Pachic Haplustoll) with 0–3% slope. Soil test results indicated soil organic matter content of 47 g kg^{−1}. This site is characterized by topography modified by glaciers and dominated by short-grass prairie with gently rolling slopes (Padbury et al., 2002).

The East sites have a history of tillage and are either certified organic land in the case of Absaraka or plots were managed following strict National Organic Program (NOP)

standards with approved amendments and pesticide inputs at Fargo. Precipitation between April and July is, on average, 244 mm (Padbury et al., 2002). Since the East sites are close in proximity and thus similar environmentally, long-term precipitation and temperature information is provided for Fargo. The long-term (1981–2010) mean annual precipitation is 498 mm and mean annual temperature is 4.4 °C (Padbury et al., 2002). The dominant soil type is a Warsing loam (fine-loamy, mixed, superactive, frigid, Oxyaquid Hapludoll) with 0–3% slope in Absaraka and Fargo Silty Clay (fine, smectitic, frigid, Epiaquert) with 0–3% slope in Fargo. This soil series is characterized by glacial till deposits located along the Red River with a level to gently sloping topography (Padbury et al., 2002). Soil test results indicated soil organic matter content of 27 and 83 g kg^{−1} in Absaraka and Fargo, respectively.

2.2 | Experimental design and treatments

This field experiment consisted of five cover crop treatments grown either under ambient rainfall conditions (dryland) or under ambient conditions with supplemental irrigation (dryland+irrigation) in 2016 and 2017 at both sites. Cover crop species were selected from a suite of several species tested in an exploratory phase of this experiment based on their germination success and relative growth rates when grown in a greenhouse under limited and erratic moisture conditions (data not reported). Treatments and crops selected were: (a) cowpea [*Vigna unguiculata* (L.) Walp.] monoculture, (b) Siberian foxtail millet [*Setaria italica* (L.) P. Beauvois] monoculture, (c) sunflower (*Helianthus annuus* L.) monoculture, (d) a 3-species warm season mixture (MIX3) of cowpea, foxtail millet, and sunflower, and (e) a 6-species warm and cool season mixture (MIX6) of cowpea, foxtail millet, sunflower, hairy vetch (*Vicia villosa* Roth), forage oat (*Avena sativa* L.), and kale (*Brassica oleraceae* L.). Treatments were randomly assigned to 6.1 by 4.6 m plots with 1.5-m borders, replicated across four blocks in a randomized complete block, 2 × 3 factorial design with 4.6 m separation between blocks at both sites. Natural Resources Conservation Service recommended seeding rates were used for each species and adjusted accordingly based on their proportion within each polyculture (Table 1).

2.3 | Management by site

At the West site, cover crops were sown using a no-till drill (Model 3P606NT, Great Plains Manufacturing, Inc.) with a 19-cm row spacing and a depth of 2.5 cm into barley (*Hordeum vulgare* L.) stubble on 23 May 2016 and 10 May 2017. Management was considered conventional, that is, non-organic, and included a spring application of

TABLE 1 Cover crop species with varieties, functional grouping classifications, and planting rates in kg pure live seed (PLS) ha⁻¹ for a field study conducted in 2016 and 2017 near Mandan, ND (West site) and Absaraka (2016) and Fargo (2017), ND (East site)

Cover crop	Variety	Functional group	kg PLS ha ⁻¹		
			Monoculture	MIX3	MIX6
Foxtail millet	Manta	WS grass	19.5	6.4	3.3
Cowpea	Iron & clay	WS legume	61.2	20.2	10.4
Sunflower	Cobalt-H	WS broadleaf ^a	4.6	1.5	0.7
Forage oat	Leggett	CS grass	—	—	12.9
Hairy vetch	vns	CS legume	—	—	4.1
Kale	Premier	CS broadleaf	—	—	0.6

Note: MIX3, 3-species cover crop mixture; MIX6, 6-species cover crop mixtures; PLS, pure live seed; WS = warm season; CS = cool season; vns, variety not stated.

^aBroadleaf cover crops are non-leguminous species.

glyphosate [N-(phosphonomethyl) glycine; 1.2 kg a.i. ha⁻¹] prior to planting and mowing of border areas between plots and replications. The West site was not fertilized during either year of the study

At the East site, cover crops were sown using a no-till drill (Model 3P606NT, Great Plains Manufacturing, Inc.) with 15.25-cm row spacing and a depth of 2.5 cm on 7 June 2016 and 9 June 2017 into recently tilled and rolled soil. For the 2016 site, the previous year was fallow with tillage; for the 2017 site, the prior crop was soybean [*Glycine max* (L.) Merr.]. Plots at the East site were managed following organic management practices. Approximately 3 d prior to planting, dry pelletized, composted poultry manure (4–3–2) was broadcast at a rate to achieve 100 kg N ha⁻¹ (Cashton Farm Supply). The composted poultry manure was incorporated using a disk cultivator.

At both sites, irrigated plots were watered using overhead sprinkler systems elevated 1.25 m above each plot with three total sprinkler heads (Rain Bird Corp.) spaced 1.0 m apart and approximately 46 cm overlapping coverage. The sprinkler system was placed diagonally so the center 7 m² of each plot was irrigated. The target irrigation amount with frequency was 12.7 mm once per week, although this varied slightly across the two sites due to rainfall and logistical constraints. A total of six and nine irrigation events occurred at the West site in 2016 and 2017, respectively, for a total of 62 and 67 mm of supplemental water. Due to well pressure issues coupled with relatively wet weather conditions overall, the East site was provided little supplemental irrigation (<10 mm) in 2016. A total of four events occurred in 2017 for a total of 51 mm of supplemental water.

2.4 | Data collection

At the West site, one 12-cm CS655 volumetric soil water content sensor (Campbell Scientific Inc.) was buried ver-

tically at 20-cm depth in one replicate of the foxtail millet single species planting, MIX3, and MIX6 treatments under dryland and dryland+irrigation conditions ($n = 3$ for each water regime) in 2016. Soil was backfilled and firmly pressed to fill holes created during burial. Sensors were used to confirm differences in soil water content were achieved with supplemental irrigation. In 2017, sensors were buried in three replicates of each single and multi-species planting, but logger failures resulted in only two replicates of data for each cover crop treatment. Data were then averaged over cover crop treatments to visually compare separation between dryland and dryland+irrigation treatments.

At the East site during 2016 and 2017, Decagon GS-1 soil volumetric water content (VWC) sensors (Decagon Devices, Inc.) were installed within a week after planting. Sensors were installed in plots on three of the four experimental blocks. A soil auger was used to bore cylindrical holes 10 cm in diameter and 30-cm deep into the center of plots. Sensor prongs were subsequently inserted into the undisturbed soil profile at 20-cm deep and soil was backfilled and firmly pressed down into the holes. Sensors were attached to Campbell CR1000X measurement and control dataloggers (Campbell Scientific Inc.). Dataloggers were programmed to log VWC once each hour, and hourly measurements were the average of measurements taken once per minute. During 2017 at the East site, one block of data was lost due to overwriting errors, so only two blocks were represented by complete datasets.

Within the center of each plot, 0.25 and 1 m² subplots at the West and East sites, respectively, were established that contained cover crops only, that is, weeds were removed regularly throughout the season. For the East sites, at full cover crop emergence (7 July 2016 and 6 July 2017) cover crop stand counts were made in these subplots. At estimated peak vegetative biomass, 14 Aug. 2016 and 9 Aug. 2017 at the West site and 22 Aug. 2016 and 16 Aug. 2017 at the East site, subplots were destructively harvested for aboveground

biomass. Biomass was separated by cover crop and weed species and dried to a constant weight at 45 °C. Water use efficiency (WUE) was calculated by dividing total dry matter production (kg ha^{-1}) by total water input which was the sum of ambient precipitation (dryland) or ambient precipitation with supplemental irrigation (dryland+irrigation) in millimeters for each water regime treatment.

Soil water balance estimates (defined here as the soil water content at planting minus the soil water content at harvest) using soil water measurements are often subtracted from inputs to estimate crop water use (St. Aime et al., 2020). Because we did not have VWC measurements for every plot, we considered water use equal to water input, and reasoned this approach would show relative if not absolute differences in WUE across cover crop treatments. To justify this approach, we used the existing VWC data to calculate soil water balances for the East site during 2016 and 2017. If cover crop treatments had no effect on these soil water balances, we reasoned that soil water balances would not affect relative differences in WUE among cover crop treatments. For the East sites, VWC was measured in three of the four experimental blocks during 2016 and two of four blocks during 2017. For these sites, we calculated soil water balances as the difference between VWC at the start of the season (shortly after planting) and VWC at harvest. Analysis of variance tests showed no cover crop mix impacts on soil water balances during either 2016 or 2017 at the East sites ($P = .26$ and $.09$, respectively, data not shown). Data for VWC were incomplete at the West site for 2016 and 2017, so statistical comparisons were not possible. Season long water balance values for cover crop treatments within one replication ranged from -0.03 to $-0.01 \text{ m}^3 \text{ m}^{-3}$ in the dryland+irrigation treatments and from 0.16 to $0.19 \text{ m}^3 \text{ m}^{-3}$ in the dryland treatments (data not shown). Observationally, these ranges of values suggest water balances differed somewhat among cover crop treatments, although not in the direction one would expect (i.e., that water content should be lower in the dryland plots). Also, because these numbers come from only one rep, we do not know the true variability associated with these numbers. Therefore, we have relied more on the data from the East site for considering crop water use equal to water input for the purpose of determining relative WUE differences among cover crop treatments.

At the East site only, soil cores were removed to determine soil macro nutrients (N–P–K) at peak cover crop vegetative growth or late July during both years. Each plot was sampled 20 times at randomly selected points in a zig-zag pattern across the plot at a depth of 0–15 cm using a soil probe with an inside diameter of 1.6 cm (AMS Inc.). The cores were mixed thoroughly to homogenize soil and analyzed for $\text{NO}_3\text{--N}$, P, and K using water extraction, the Olson procedure, and 1 N ammonium acetate methods, respectively.

2.5 | Data analysis

Analysis of variance tests were conducted using the PROC MIXED procedure in SAS version 9.4 (SAS Institute). Cover crop treatment, water regime (dryland vs. dryland+irrigation) and year were considered fixed effects and replication a random effect. Prior to testing, data were assessed for normality (PROC UNIVARIATE) and variance homogeneity (PROC GLM, Levene's test). No data transformations were necessary. When F tests indicated an interaction, the "slice" option was used in the LSMEANS statement using a Bonferroni correction for multiple comparisons. Preliminary analyses of aboveground biomass production and WUE resulted in significant site, site $\times$ year, and site $\times$ cover crop treatment interactions. Thus, sites were analyzed separately. For proportional contributions of each cover crop species to total production in MIX3 and MIX6, water regime was also analyzed separately. Differences among means were tested using Tukey's method and an alpha level of .05 for all F tests and means separations. SAS PROC CORR was used to estimate correlation between total dry cover crop biomass and soil $\text{NO}_3\text{--N}$ content (measured at peak vegetative biomass).

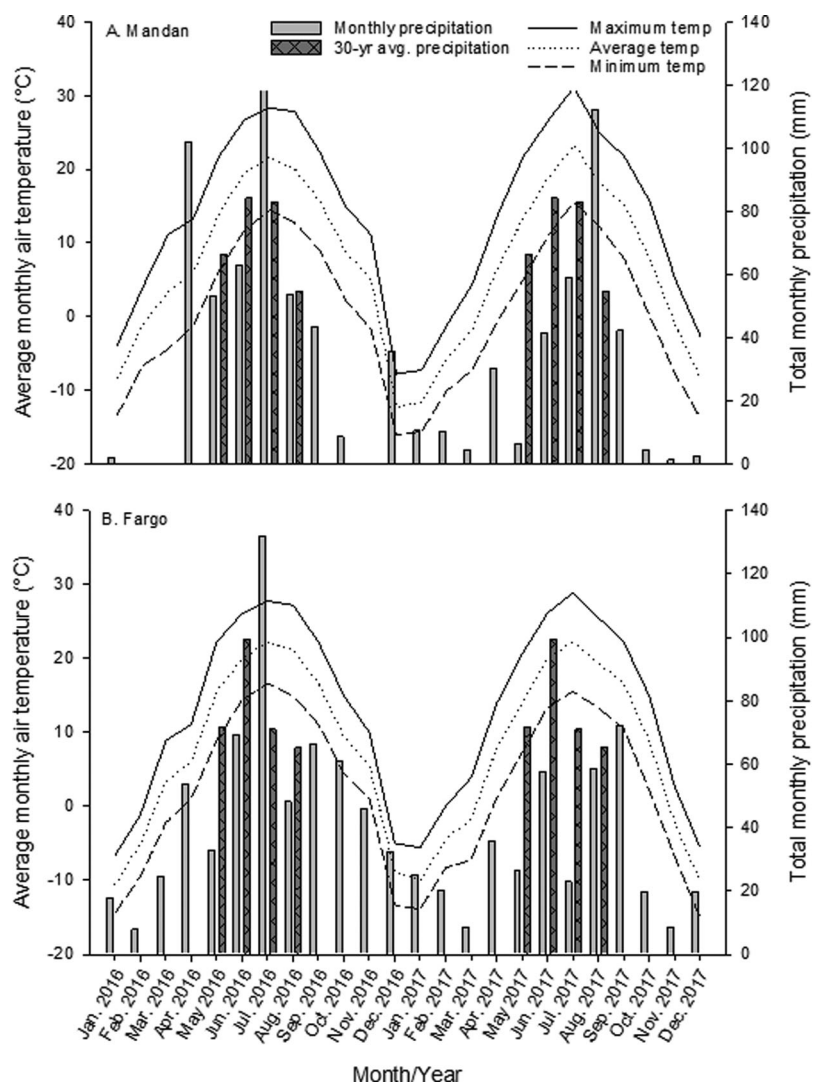
To evaluate cover crop production variability, an analysis of coefficient of variation (CV) was used. Coefficient of variation was calculated by taking the mean and standard deviation of aboveground biomass production expressed as a dry weight for each cover crop treatment across water regimes and year within a site ($n = 16$) and then pooled across sites ($n = 32$). The CVs were used to analyze variability in functional group proportional contributions to total biomass production across water regimes to evaluate changes to altered moisture conditions then across water regimes and cover crop treatment to assess the contribution stability of each functional group to overall productivity.

3 | RESULTS

3.1 | Precipitation and volumetric water content

Growing season precipitation was above average at the West site in 2016 which followed a wet spring of more than 100 mm of precipitation in April alone (Figure 1a). In 2017, growing season precipitation was below average and a dry May, when planting occurred, was preceded by 20 mm of precipitation in April (Figure 1a). However, as the season progressed monthly precipitation increased to twice the 30-yr (1981–2010) average around termination in August 2017 (Figure 1a). At the East site, 2016 was also the wetter of the 2 yr, but still below the 30-yr (1981–2010) season total average, particularly around planting time in early June and

FIGURE 1 Monthly maximum (solid lines), average (dotted lines), and minimum (dashed lines) temperatures, and total monthly (rainfall + snowmelt; solid bars) and growing season (May–August) 30-yr (1981–2010) average monthly precipitation (hashed bars) for (A) Mandan, ND and (B) Fargo, ND for 2016 and 2017. Weather data obtained from North Dakota Agricultural Weather Network (<https://ndawn.ndsu.nodak.edu/index.html>; accessed 8 Dec. 2020) and the National Climatic Data Center (NOAA, <https://www.ncdc.noaa.gov/cdo-web/>; accessed 8 Dec. 2020). Planting at the Mandan site occurred 23 May 2016 and 10 May 2017 and termination occurred 14 Aug. 2016 and 9 Aug. 2017. Planting at the Absaraka/Fargo sites occurred 7 June 2016 and 9 June 2017 and termination occurred 22 Aug. 2016 and 16 Aug. 2017



at termination in August (Figure 1b). This site received more consistent precipitation in the months leading up to planting both years of the study (Figure 1b). Total growing season water input in the form of precipitation alone or precipitation plus supplemental irrigation ranged from 354 mm at the West site to 165 mm at the East site. This allowed for a broad range of moisture conditions in which to compare single species and multi-species cover crop biomass production, production stability, WUE, and proportional contribution of species to total production. Temperatures for the two growing seasons of the study ranged from 5.0 to 31.1 °C in Mandan (Figure 1a) and from 7.2 to 28.9 °C in Fargo (Figure 1b).

Volumetric water content data showed little difference between water regime treatments as monitored at the 20-cm depth in Absaraka in 2016, which was expected given very little supplemental irrigation was provided (Figure 2). In Fargo in 2017, slight differences in VWC were apparent, particularly toward the end of the growing season with greater VWC values in the dryland+irrigation treatment. Differences were more apparent at the West site (Figure 2), which

is typically the more arid of the two sites (Padbury et al., 2002). In 2016, dryland+irrigation treatments had consistently higher VWC values. In 2017, values were highly variable for the dryland+irrigation treatment as compared to dryland, but VWC values then began to increase considerably by the 2nd week of July. Volumetric water content values were much more variable at the West site as compared to the East site, with steeper declines in dry-down following a precipitation or irrigation event. This could possibly be attributed to the higher clay content soils dominating the East site, which are characterized by greater water retention and, thus, VWC values at the East site did not exhibit the variability over time as the West site. This may also imply that surface runoff occurred at the East site where VWC was already high when precipitation occurred, even though similar amounts were received (using the precipitation event in July 2016 as an example) at both sites, thereby resulting in no soil water recharge and less variable VWC values as compared to the West site. Alternatively, higher temperatures and lower humidity at the West site may have led to more evaporative water losses resulting

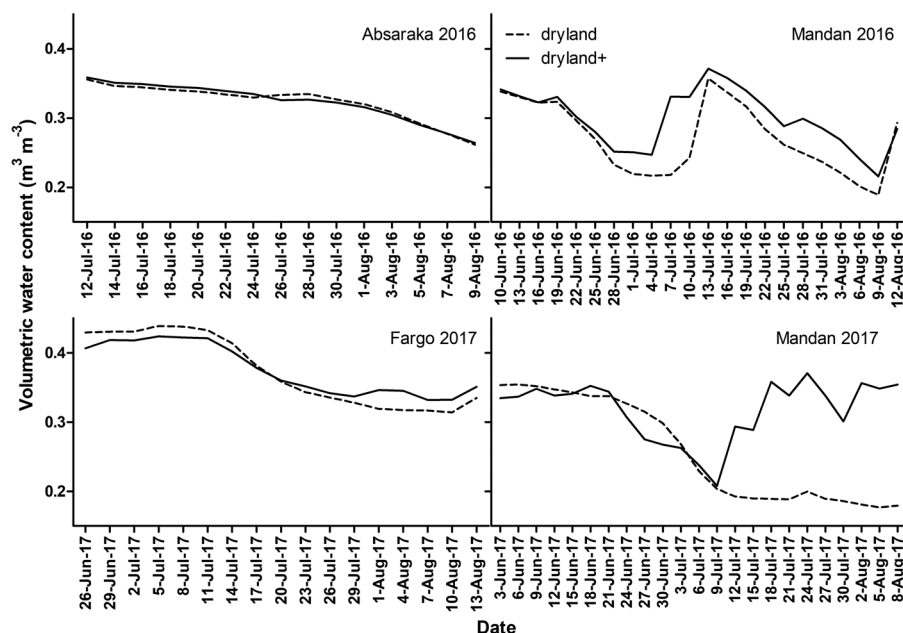


FIGURE 2 Volumetric water content (VWC, $\text{m}^3 \text{m}^{-3}$) measured at a depth of 20 cm in Absaraka (2016) and Fargo (2017), ND ($n = 3$ and 2 for each water regime/site, respectively) and Mandan, ND ($n = 3$ for each water regime). The dryland treatment (dashed lines) reflects ambient precipitation only while the dryland+irrigation treatment (solid lines) reflects ambient precipitation with supplemental irrigation

in more variable soil surface moisture. As a result of soil type differences, the West site had much lower VWC values during dry periods than the East site. Although rainfall totals were not dissimilar between sites, and at times lower rainfall was received at the East site, soils dominating the East site resulted in consistently greater soil water content.

3.2 | Cover crop biomass production

Differences between sites and the site $\times$ cover crop treatment interaction were driven by greater foxtail millet production at the West site compared to the East site and greater cowpea production at the East site compared to the West site (Table 2). At the West site, the main effect of cover crop had a significant effect on biomass production ($P < .001$), but there was no effect of water regime or year (means not shown). All other cover crop treatments produced significantly more biomass (at least 75% more) than cowpea grown in monoculture (Table 2). Averaged over years, the most productive cover crop treatment, sunflower monoculture, produced $11,295 \text{ kg ha}^{-1}$ of aboveground biomass compared to $2,146 \text{ kg ha}^{-1}$ for cowpea grown in monoculture. There was no difference in biomass production among foxtail millet, sunflower, MIX3, and MIX6 (Table 2).

At the East site, the main effects of year and cover crop treatment were significant ($P < .001$), but there was no effect of water regime and no significant interactions (Table 2). Total production averaged across cover crop treatments in 2016 was $11,175 \text{ kg ha}^{-1}$ compared to $7,829 \text{ kg ha}^{-1}$ in 2017,

but it is unclear if these differences are strictly due to site differences between Absaraka and Fargo or other factors. Poor foxtail millet emergence at the Fargo site may have been a significant contributing factor to reduced productivity in 2017. Based on cover crop stand counts made in early July, foxtail millet stand counts at Fargo during 2017 were 36% lower than Absaraka stands in 2016. Consequently, at Fargo during 2017, foxtail millet grown in monoculture produced significantly less biomass than all other cover crop treatments; 62% less than the most productive treatment (Table 2). Averaged over year, MIX3 was numerically the most productive cover crop treatment, $11,793 \text{ kg ha}^{-1}$, but similar to cowpea, sunflower, and MIX6.

3.3 | Cover crop production stability

Coefficient of variation analyses indicated that MIX6 had the lowest variability in aboveground biomass yield at the West site when pooled across years and water regime treatments (25%) as compared to all other treatments, which were all above 49% (Table 2). Cowpea had the highest CV with 58%. At the East site, cowpea had the lowest CV with 24% and millet the highest with 39% (Table 2). However, when pooled across sites, MIX6 had the lowest yield variability with 29% compared with 43, 44, 59, and 73% for MIX3, sunflower, millet, and cowpea. Although cowpea performed well at the East site, possibly due to greater soil water content at those sites, it was the poorest performing monoculture at the West site. This led to a high overall CV when pooled across sites.

TABLE 2 Aboveground biomass yield (kg ha^{-1}) of cover crop single species and multi-species plantings under dryland (ambient rainfall) and dryland+irrigation (ambient rainfall with supplemental irrigation) in a field study conducted in 2016 and 2017 near Mandan, ND (West site) and Absaraka (2016) and Fargo (2017), ND (East site)

Site-year	Foxtail millet	Cowpea	Sunflower	MIX3	MIX6
Dryland, dry biomass yield, kg ha^{-1}					
West					
2016	6,443	2,224	9,933	8,779	9,117
2017	9,826	1,706	11,309	9,339	8,795
Dryland+irrigation, dry biomass yield, kg ha^{-1}					
2016	6,497	3,502	12,614	9,608	11,529
2017	13,241	1,153	11,325	7,046	11,409
2-yr mean	9,002 ^a	2,146 ^b	11,295 ^a	8,693 ^a	10,212 ^a
CV ^b	49	58	50	50	25
Dryland, dry biomass yield, kg ha^{-1}					
East ^c					
2016	5,762	12,535	11,004	13,873	11,726
2017	3,243	7,680	6,876	10,761	8,689
Dryland+irrigation, dry biomass yield, kg ha^{-1}					
2016	4,412	11,921	12,704	15,052	12,764
2017	4,605	10,886	9,462	7,486	8,603
2016 mean	5,087 ^b	12,228 ^a	11,854 ^a	14,462 ^a	12,245 ^a
2017 mean	3,924 ^b	9,283 ^a	8,169 ^a	9,124 ^a	8,646 ^a
2-yr mean	4,505	10,755	10,011	11,793	10,446
CV ^b	39	24	35	36	34
Overall CV ^d	59	73	44	43	29

Note: MIX3, 3-species cover crop mixture; MIX6, 6-species cover crop mixtures.

^aDifferent lower case letters indicate statistical differences ($P < .05$) among cover crop treatments averaged over years and water regime treatments for the West site and over water regimes within years for the East site.

^bCoefficient of variation, expressed as a percentage, of aboveground biomass yield for each cover crop treatment pooled across water regime treatment, replicates, and years within a site ($n = 16$).

^cDryland+irrigation treatments received minimal supplemental irrigation in 2016.

^dCoefficient of variation, expressed as percentage, of aboveground biomass yield for each cover crop treatment pooled across sites, water regime treatment, replicates, and years ($n = 32$).

Although MIX6 resulted in greater yield stability, that is, low variability, at the West site, it was not the most stable cover crop treatment at the East site. The CV value for MIX6 at the East site was slightly lower than all other cover crop treatments except for the cowpea single species planting.

3.4 | Cover crop proportional contribution

While highly variable for both sites, variability of proportional contribution by cover crop species results were lower for the West site (Table 3). At the West site, Siberian foxtail millet, a warm season grass, contributed more to overall MIX3 production both years and within each water regime than did cowpea, a warm season legume. In 2016, however, there were no significant differences between sunflower, a warm season broadleaf, and foxtail millet, a warm season grass, in MIX3 (Table 3). In the MIX6 cover crop treatment,

forage oat contributed greater biomass to overall production than other component cover crops in 2016 (65% in dryland and 72% in dryland+irrigation treatments; Table 3). In 2017, foxtail millet dominated in the MIX3 treatment and contributed 96% to overall productivity when averaged across water regime (Table 3). In the MIX6 treatment in 2017, forage oat again contributed greater to overall production, but statistical differences were not detected between forage oat, foxtail millet, and kale in the dryland treatment ($P = .001$) and forage oat and foxtail millet in the dryland+irrigation treatment ($P < .001$). Again in 2017 when conditions were drier, kale contribution to overall production increased, particularly in the dryland treatment. Overall, variability of proportional contribution from each cover crop species at the West site showed that the grasses had the lowest CV values, 38% for foxtail millet and 39% for forage oat (Table 3). All other cover crops had CV values above 96% when pooled across years due to the high response variability among

TABLE 3 Cover crop proportional contribution by species to total production in 3-species cover crop mixture (MIX3) and 6-species cover crop mixtures (MIX6) cover crop polycultures at the West (Mandan, ND) site in 2016 and 2017. Mean proportional contribution by cover crop species averaged across water regime, across water regime and years, and coefficients of variation (CV, %) are also presented

	2016			2017				
Cover crop species	Dryland	Dryland + irrigation	Mean	Dryland	Dryland + irrigation	Mean	Overall mean	CV ^a
MIX3								
Foxtail millet	0.55a ^b	0.57a	0.56	0.98A	0.93A	0.96	0.76	38
Cowpea	0.18b	0.13b	0.16	0.01B	0.06B	0.04	0.10	141
Sunflower	0.27ab	0.30a	0.29	0.01B	0.01B	0.01	0.15	116
MIX6								
Foxtail millet	0.20b	0.14b	0.17	0.19AB	0.21AB	0.20	0.19	62
Cowpea	0.07c	0.04bc	0.06	0.01B	0.01B	0.01	0.04	96
Sunflower	0.01c	0.01c	0.01	0.17B	0.09B	0.13	0.07	221
Forage oat	0.65a	0.72a	0.69	0.35A	0.53A	0.44	0.57	39
Hairy vetch	0.04c	0.04c	0.04	0.02B	0.01B	0.02	0.03	129
Kale	0.03c	0.04c	0.04	0.26A	0.16B	0.21	0.13	107

^aCoefficient of variation (CV) of fractional contribution of each functional group to total aboveground biomass yield pooled across water regime treatment and replicates ($n = 16$).

^bDifferent lower case (2016) and upper case (2017) letters indicate statistical differences ($P < .05$) among component cover crops within a cover crop mixture and water regime treatment.

TABLE 4 Cover crop proportional contribution by species to total production in 3-species cover crop mixture (MIX3) and 6-species cover crop mixtures (MIX6) cover crop polycultures at the East (Absaraka and Fargo, North Dakota) site in 2016 and 2017. Mean proportional contribution by cover crop species averaged across water regime, across water regime and years, and coefficients of variation (CV, %) are also presented

	2016			2017				
Cover crop species	Dryland	Dryland + irrigation	Mean	Dryland	Dryland + irrigation	Mean	Overall mean	CV ^a
MIX3								%
Foxtail millet	0.74a ^b	0.65a	0.69	0.38	0.58	0.48	0.59	38
Cowpea	0.01c	0.03c	0.02	0.08	0.15	0.12	0.07	142
Sunflower	0.25b	0.32b	0.29	0.54	0.27	0.41	0.34	116
MIX6								
Foxtail millet	0.69a	0.69a	0.69	0.15BC	0.07BC	0.11	0.40	77
Cowpea	0.03c	0.05c	0.04	0.01C	0.01C	0.01	0.02	101
Sunflower	0.24b	0.25b	0.25	0.12bC	0.20B	0.16	0.21	66
Forage oat	0.03c	0.01c	0.02	0.43A	0.44A	0.43	0.23	97
Hairy vetch	0.00c	0.00c	0.00	0.01C	0.10BC	0.06	0.03	331
Kale	0.01c	0.00c	0.00	0.28AB	0.18BC	0.23	0.11	126

^aCoefficient of variation (CV) of fractional contribution of each functional group to total aboveground biomass yield pooled across water regime treatment and replicates ($n = 16$).

^bDifferent lower case (2016) and upper case (2017) letters indicate statistical differences ($P < .05$) among component cover crops within a cover crop mixture and water regime treatment.

replicates within a year and water regime treatment (data not shown) and variability between years.

At the East site in the MIX3 cover crop treatment, foxtail millet also contributed greater biomass to overall production along with sunflower than cowpea; however, treat-

ment differences were only detected in 2016 (Table 4). In the MIX6 treatment, foxtail millet was the dominant crop in dryland (69%) and dryland+irrigation (69%) treatments in 2016 (Table 4). In 2017, which was also the drier year at this site, forage oat contributed significantly more to overall

TABLE 5 Water use efficiency (WUE; $\text{kg ha}^{-1} \text{mm}^{-1}$) of cover crop single species and multi-species plantings under dryland (ambient rainfall) and dryland+irrigation (ambient rainfall with supplemental irrigation) in a field study conducted in 2016 and 2017 near Mandan, ND (West site) and Absaraka (2016) and Fargo (2017), ND (East site). Mean WUE ($\text{kg ha}^{-1} \text{mm}^{-1}$) averaged across regime are also presented

Site-year	Foxtail millet	Cowpea	Sunflower	MIX3	MIX6
Dryland, WUE, $\text{kg ha}^{-1} \text{mm}^{-1}$					
West					
2016	18	6	28	25	26
2017	35	6	40	33	31
Dryland+irrigation, WUE, $\text{kg ha}^{-1} \text{mm}^{-1}$					
2016	18	10	36	27	33
2017	47	4	40	26.2	41
2-yr mean	30a ^a	7b	36a	28a	33a
Dryland, WUE, $\text{kg ha}^{-1} \text{mm}^{-1}$					
East ^b					
2016	25	54	47	59	50
2017	16	36	32	50	40
Dryland+irrigation, WUE, $\text{kg ha}^{-1} \text{mm}^{-1}$					
2016	19	51	54	64	55
2017	21	51	44	35	40
2016 mean	22b	52a	51a	62a	52a
2017 mean	18b	43a	38a	43a	40a
2-yr mean	20	48	44	52	46

Note: MIX3, 3-species cover crop mixture; MIX6, 6-species cover crop mixtures.

^aDifferent lower case letters indicate statistical differences ($P < .05$) among cover crop treatments averaged over years and water regime treatments for the West site and over water regimes within years for the East site.

^bDryland+irrigation treatments received minimal supplemental irrigation in 2016.

production in the dryland (43%) and dryland+irrigation (44%) water regime treatments, but not significantly more than kale in the dryland treatment at 28% (Table 4). Kale contribution increased in 2017 in both treatments, statistically similarly to forage oat in the dryland treatment. Like the West site, foxtail millet had low variability averaged over years in MIX3 ($\text{CV} = 38\%$) at the East site, but not lower than sunflower in MIX6 ($\text{CV} = 66\%$ for sunflower and 77% for foxtail millet). Unlike the West site, forage oat did not have a consistent contribution to overall biomass at the East site due to its poor performance in 2016 under wetter conditions. Cowpea and hairy vetch, both legumes, had particularly high CV values, especially in the MIX6 cover crop treatment due to a higher number of 0 values for biomass production and showed low contributions to overall production at both sites overall.

3.5 | Cover crop water use efficiency

At the West site, the main effect of cover crop treatment had a significant ($P < .001$) effect on WUE, but there was no effect of water regime. Averaged across years and water regimes, the cowpea single species planting produced the least amount of aboveground biomass per unit of water ($7 \text{ kg ha}^{-1} \text{mm}^{-1}$)

as compared to all other cover crop treatments (Table 5). Values for foxtail millet, sunflower, MIX3, and MIX6 varied between 28 and $36 \text{ kg ha}^{-1} \text{mm}^{-1}$ but were not significantly different from each other.

Similarly, at the East site no water regime effect was detected. Differences among cover crop treatments ($P < .001$) and between years ($P = .003$) were, however, detected with year differences, again, possibly explained by differences in locations and crop establishment. Following biomass yield findings, WUE values were greater in 2016 when more rainfall was received than in 2017 at this site (Table 5). Foxtail millet produced the least amount of biomass per unit of water input ($20 \text{ kg ha}^{-1} \text{mm}^{-1}$) than all other cover crop treatments and values ranged between 44 and $52 \text{ kg ha}^{-1} \text{mm}^{-1}$ for those treatments.

3.6 | Soil nutrients

In 2016, soil analyses (determined at peak vegetative cover crop development) from the East (Absaraka) site suggested that supplemental irrigation did not influence soil $\text{NO}_3\text{-N}$, but cover crop treatments were associated with differences in soil $\text{NO}_3\text{-N}$ ($P = .012$, Table 6). However, analyses conducted to

TABLE 6 Mean soil nitrogen ($\text{NO}_3\text{-N}$, kg ha^{-1}), phosphorus (P, parts per million), and potassium (K, parts per million) measured at 0-to-15-cm depth for dryland (ambient rainfall) and dryland+irrigation treatments (ambient rainfall with supplemental irrigation) and cover crop treatment (cowpea monoculture, foxtail millet monoculture, sunflower monoculture, 3-species mixture [MIX3], and 6-species mixture [MIX6]) measured at the East site in 2016 (Absaraka, ND) and 2017 (Fargo, ND)

Treatment	East 2016 ^a			East 2017 ^a		
	$\text{NO}_3\text{-N}$	P	K	$\text{NO}_3\text{-N}$	P	K
	kg ha^{-1}	ppm		kg ha^{-1}	ppm	
Dryland	127A	14A	142A	197A	31A	490A
Dryland+irrigation	123A	13A	137A	163B	26B	437B
Cowpea	138a	12a	141a	169a	27a	446a
Foxtail millet	127ab	12a	143a	197a	29a	460a
Sunflower	143a	15a	143a	182a	27a	445a
MIX3	102b	14a	133a	178a	29a	489a
MIX6	117ab	14a	140a	177a	33a	476a

^aMeans separation ($P < .05$) for each nutrient within each site-year denoted by differing uppercase letters for comparisons between water regime treatments and by differing lowercase letters within cover crop treatments.

determine correlations between total dry cover crop biomass and soil $\text{NO}_3\text{-N}$ produced correlation coefficients of -0.24 ($P = .1373$) and 0.08 ($P = .6158$) for the East 2016 (Absaraka) and East 2017 (Fargo) sites, respectively. In contrast to $\text{NO}_3\text{-N}$, soil P and K were not influenced by either water regime or cover crop treatments at the Absaraka site during 2016. Soil P and K were also not correlated with total cover crop biomass (data not shown). Unlike 2016 at Absaraka, cover crops did not influence soil $\text{NO}_3\text{-N}$, P, or K during 2017 at Fargo (Table 6). However, supplemental irrigation was associated with reduced soil $\text{NO}_3\text{-N}$ ($P = .008$), P ($P = .0361$), and K ($P = .0288$) compared to dryland treatments (Table 6).

4 | DISCUSSION

Our finding that cover crop mixtures do not out-yield monocultures, with some exceptions, does not support our hypothesis and is consistent with late-summer planting information collected by Sanderson et al. (2018) as well as findings by Nielsen et al. (2015a). These findings are contrary to what Sanderson et al. (2018) found for spring-planted cover crops, however, and what others have reported whereby the average of mixtures was found to out-yield the average of monocultures (Khan & McVay, 2019; Sanderson et al., 2018; Smith et al., 2014; Wortman et al., 2012). However, several of these reports compare production averages between mixtures and monocultures, which are not reported in our findings. We felt it was important to evaluate single species plantings individually to assess their performance and versatility (via CV analysis) across the two differing sites. In addition, it has been suggested that lower producing monocultures may suppress average monoculture productivity (Florence et al., 2019), resulting in no productivity difference between mixture and the average of monocultures. Moreover, lack of

consistent results regarding productivity of cover crop mixtures vs. monocultures might be expected when considering ecological theory underpinning overyielding. Transgressive overyielding (i.e., a mixture is more productive compared to the most productive component in monoculture) is expected to occur when various species in an assemblage occupy different niches (i.e., niche differentiation leads to co-existence, which maximizes overall productivity by reducing competition). Conversely, niche overlap leads to competition, which would negate transgressive overyielding (Ashton et al., 2010; Florence et al., 2019; Schmid et al., 2008). Because some cover crop mixtures may result in greater niche differentiation than others, lack of consistent results regarding overyielding should not be surprising. We chose the species in our mixtures based on their ability to tolerate dry conditions when grown in monocultures, not based on known niche differentiation, which may be one reason why we did not measure transgressive overyielding (Wendling et al., 2017).

Greater overall productivity at the East site may be partially explained by more consistent precipitation during the growing season and in the months leading up to planting, especially during 2016, as well as soil differences discussed more below, as compared to the West site (Figure 2a and 2b). Because cover crop biomass production was relatively high, up to $15,052 \text{ kg ha}^{-1}$, considering water uptake by cover crops that otherwise would be available for subsequent cash crops is critical. This is particularly true if cover crops are incorporated as full-season plantings in crop rotations grown in semi-arid regions such as the NGP. Although we did not evaluate crop yield in our study, other studies have shown that spring-planted cover crops may depress subsequent cash crop yields (Nielsen & Vigil, 2005; Unger et al., 2006), while others have shown no effect, either positive or negative, on subsequent crop yield (Nielsen et al., 2016). In comparing water uptake by multi-species mixtures vs. single species

plantings, Khan and McVay (2019) and Nielsen et al. (2015b) found no differences. Collectively, this suggests multi-species mixtures may not adversely impact subsequent crop yields more than a single species cover crop, although this requires further investigation. Further, soil water recharge from winter precipitation may counterbalance water uptake by cover crops grown the season before a cash crop (Khan & McVay, 2019). Winter precipitation at both sites was substantial (Figure 2a and 2b), and particularly consistent at the East site, and may have been enough to restore depleted soil water. This was not a focus of our evaluations, however, but should be an important consideration for future work in semi-arid climates.

Supporting our hypothesis, the low overall variability across sites suggests that more complex multi-species mixtures are more consistent in production under variable rainfall and, also, more versatile when accounting for variability in site differences such as climate, soil, and management types. Wortman et al. (2012) observed lower CV values for the average of cover crop mixtures as compared to the average of monocultures but found that these differences were not statistically different. Similarly, Khan and McVay (2019) found no differences in stability between monocultures and mixtures but reported a low CV value for the legume cover crop, in this case yellow dry pea (*Pisum sativum* L.), which we observed only at the East site with cowpea.

Our findings, however, support classic plant ecology studies conducted in perennial grassland ecosystems that were the first to report a positive correlation between ecosystem diversity and plant community stability, or what is referred to as the diversity–stability relationship (Frank & McNaughton, 1991; Schlapfer & Schmid, 1999; Tilman, 1996; Tilman & Downing, 1994). Species richness, that is, the number of species, is believed to increase stability, measured by the resistance to change to disturbances such as drought (Frank & McNaughton, 1991), through differential species' response to those disturbances, such that the sum of those responses results in more stable community dynamics over the long term (Diaz & Cabido, 2001; McCann, 2000) as compared to simplified communities or monocultures. Subsequent experiments provided evidence to support the idea that the number of species alone does not result in more stable communities, but that functional composition and diversity, or the type and range of functional traits of the organisms present in an ecosystem (Diaz & Cabido, 2001), are stronger determinants of stability and overall functioning of an ecosystem (Hooper & Vitousek, 1997; Tilman et al., 1997). In this experiment, the 6-species mixture, which contained the greatest species richness as well as the most functionally diverse assemblage of species (grasses, nitrogen-fixing legumes, non-leguminous species, warm and cool season species, varying plant architectures and leaf traits, varying rooting depths, etc.), resulted in the most stable yields across sites and years.

Finally, CV values were generally lower at the East site as compared to the West site, which could be attributed to soil type differences, more consistent precipitation (Figure 1a and 1b), likely resulting in the observed lower variation in VWC (Figure 2). The CVs reported from this study represent cover crop treatment response for just 2 yr at two sites, and a larger, more robust dataset capturing more years and from more diverse environments would provide greater confidence regarding conclusions drawn from comparisons between single cover crop plantings and cover crop mixtures.

Collectively from both sites, the increase in kale production during the dry year and under dryland conditions indicates kale does well in drier conditions and may be a good choice for inclusion in cover crop mixtures in drier climates. Further, kale and other *Brassica* cover crops have been found to reduce soilborne pathogens, reduce weed pressure, increase N uptake following a cash crop, and can also produce quality forage for livestock, thus providing a suite of additional ecosystem service benefits (Chakwizira et al., 2014; Haramoto & Gallandt, 2004, 2005; Janosevic et al., 2017; Monfort et al., 2007; Seigies & Pritts, 2006). Thus, future studies should investigate its performance in single species as well as multi-species cover crop plantings.

Nielsen et al. (2015a) and Khan and McVay (2019) also showed high proportional contribution from grass species and low contributions from legume species as was observed at both sites in this study. However, even at low proportional contributions to total mixture productivity, the inclusion of leguminous and non-leguminous broadleaf cover crops may provide added benefits such as enhanced forage quality (Nielsen et al., 2015a; Sanderson et al., 2018), production of high-quality residues (Snapp et al., 2005), and supply of inorganic N to the subsequent cash crop (Finney & Kaye, 2017). In addition, when combined with grasses, legumes may accumulate more N than if grown alone due to the N scavenging abilities of non-leguminous species (Creamer & Bennett et al., 1997), resulting in an increase in biological N fixation by legumes (Ofori & Stern, 1987), and in resource sharing (or facilitation) through the transfer of N by legume roots to non-leguminous species (Giller et al., 1991).

Forage oat, a cool season grass, has been found to be competitive when included in mixtures (Wendling et al., 2017). This is at least partially due to a high root length density at deeper soil layers (Wendling et al., 2016), which allows it to explore more soil resources and be productive during times of limited precipitation. Foxtail millet and millet species in general are considered drought tolerant (Baltensperger, 1996; Nadeem et al., 2020; Nematpour et al., 2019) due to their highly efficient C4 photosynthetic pathway (Matsuura & An, 2020). Therefore, it is possible that the two grass cover crops used in this study perform better under limited water conditions, thereby producing with more consistency, which appears to be the case for the West site.

It is also plausible that since grass species typically have faster growth rates (Baraibar et al., 2018) and legumes have slower growth rates (Snapp et al., 2005), oat and millet were able to gain a competitive advantage over other cover crop species, legumes in particular, thereby contributing to overall productivity more consistently than other functional types.

Nielsen et al. (2015a) and Nielsen et al. (2015b) used a different metric for calculating WUE, and they also reported no differences in water use and water use efficiency between cover crop mixtures and single species plantings. For this analysis, we calculated WUE using the ratio of biomass accumulation to total water input to the system (Sinclair et al., 1984). This approach is supported by Hendrickson et al. (2013) who attributed biomass accumulation, not season long water balances, as the primary driver of WUE. However, this analysis assumes total water input equates to plant-available water, which may not be the case. A more indicative measure of WUE would require using the standardized ASCE equation (Allen et al., 2005) to calculate daily reference evapotranspiration. This requires determination of crop evapotranspiration (ET_c ; Moroke et al., 2011), which is challenging to determine in mixtures. Further, we were unable to calculate water use with adequate replication due to loss of sensor data. Measurements of stomatal conductance (a proxy for water stress) in the three warm season species used in this study, cowpea, sunflower, foxtail millet, when grown as single species and in the two mixtures did not show a clear trend in crop water stress (data not shown), unlike other studies that have observed reduced stress in some component species in multi-species systems (Franco et al., 2018). This, along with no differences in season long water balances among treatments we reported in the materials and methods section, further reinforces our findings that no differences were detected among cover crop treatments or between water regimes as related to water use or water use efficiency.

Overall, the lack of a significant water regime effect for biomass yield and WUE could indicate our results either (a) hold over a variety of precipitation regimes (from a seasonal minimum of 214 mm [precipitation] to a maximum of 354 mm [precipitation + irrigation] at the West site and 165 mm [precipitation] to 244 mm [precipitation + irrigation] at the East site) or (b) that water impacts were not detected due to insufficient supplemental irrigation and dry conditions to result in a detectable difference. Since no water regime effect was detected either year at the West site where we had a wet year followed by a dry year, we believe the former to be true. Differences in VWC curves between the two water regime treatments indicates a positive soil response to supplemental irrigation. However, differences in both soil and plant responses may not have been significant enough to result in the statistical detection of treatment differences. In one of the few other studies comparing cover crop mixture and monoculture production and water use in a semi-arid environment

with varying water inputs, Nielsen et al. (2015a) reported mixed results when dryland and irrigated treatments were compared in the central Great Plains, with some sites producing more biomass and greater WUE under irrigation and others reporting similar values. Evaluation sites used in their study, however, received much less precipitation than sites in this study (71–178 mm vs. 165–214 mm). Thus, greater plant response may have resulted from irrigation in drier environments used in their study. Further, differences in VWC curves between the two water regime treatments indicates a positive soil response to supplemental irrigation. On the other hand, differences in both soil and plant responses may not have been significant enough to result in the statistical detection of treatment differences and it is also plausible that biomass yield was limited by other factors other than soil water content.

Cover crop treatments influenced soil $\text{NO}_3\text{-N}$ at Fargo in 2017 but not at Absaraka in 2016 (Table 6). One explanation for this difference is the generally higher nutrient levels at the Fargo site, because of a recent history of synthetic fertilizer inputs, as well as N credits from the soybean crop grown on that field the previous year. Another factor may have been poor emergence of foxtail millet (64 plants m^{-2} in 2017 compared to 100 plants m^{-2} in 2016) because of early season drought in 2017 followed by a large rainfall event (34.6 mm on 13 June, data not shown) that caused soil crusting, which produced less competitive and robust cover crop stands (especially for smaller-seeded foxtail millet), thereby using less N. With larger amounts of most nutrients present at Fargo, neither intra- nor inter-specific competition for nutrients was likely as intense as at Absaraka, the site with a longer history of certified organic management with fewer fertilizer inputs and correspondingly lower levels of macronutrients (Table 6) coupled with much more dense cover crop stands.

At Absaraka during 2016, soil in treatments planted to cowpea and sunflower monocultures contained greater soil $\text{NO}_3\text{-N}$ than those planted to MIX3 (138 and 143, respectively, vs. 102 kg ha^{-1} N). Because cowpea is a nitrogen-fixing legume, it is expected that greater N would be associated with the cowpea monoculture in contrast to the MIX3 mixture where cowpea biomass made up only 2% of MIX3 biomass. Sunflower may not have used much N at the depth we measured soil nutrients (15 cm), because this crop can mine N from soil depths as great as 1.2 m (Franzen, 2016). Soil in treatments planted to MIX6 and foxtail millet contained intermediate amounts of $\text{NO}_3\text{-N}$, 117 and 127 kg ha^{-1} N, respectively. MIX6 contained two legumes, warm-season cowpea and cool-season hairy vetch, whereas MIX3 contained only the warm-season legume (cowpea). Vetch was not present in any plots during 2016, but the cowpea represented 4% of the total biomass. However, this slight difference did not result in $\text{NO}_3\text{-N}$ differences between the two mixtures.

Cover crop total biomass production did not correlate with soil $\text{NO}_3\text{-N}$ differences, and all the cover crop treatments

produced a similar amount of biomass, except for foxtail millet monoculture, which produced less than half the amount of biomass than the other cover crops (Table 2). The amount of $\text{NO}_3\text{-N}$ left in the soil at peak vegetative growth (when soil nutrients were sampled) seems relatively high, with values ranging from 102 to 138 kg ha^{-1} for East 2016 (Absaraka) and from 169 to 197 kg ha^{-1} for East 2017 (Fargo). Both sites were fertilized with composted poultry manure, from which N mineralizes slowly over time (Tyson & Cabrera, 1993), which could explain higher than expected soil N values measured near the end of the growing season. The slightly greater soil $\text{NO}_3\text{-N}$ levels measured at Fargo during 2017 compared to Absaraka in 2016 are probably the result of N credits from the previous year's (2016) crop of soybean. Additionally, the more finely textured clayey soil at Fargo would retain more $\text{NO}_3\text{-N}$ than the sandier soil at Absaraka.

Foxtail millet is an effective soil N scavenger (Finney et al., 2016), which may explain why the foxtail millet monoculture reduced soil $\text{NO}_3\text{-N}$ to a comparable level as both MIX 3 and MIX6 at the East 2016 site (Table 6), even though MIX 3 and MIX6 each produced more than twice the biomass of foxtail millet grown alone at that site-year (14,462 and 12,245, respectively, vs. 5,087 kg ha^{-1} , Table 2). Niche complementarity (Ashton et al., 2010) may also explain why the MIX3 treatment depleted soil N to the lowest level of all the cover crops, especially given that MIX3 foxtail millet biomass exceeded that of the monoculture. If each component crop (i.e., sunflower, cowpea, and foxtail millet) occupied different rooting zones, then presumably intraspecific competition for N may have been reduced, thus allowing each component to use proportionally more N compared to each component grown in monoculture. However, any possible niche differentiation leading to complementary N uptake did not lead to increased cover crop mixture biomass. Similar results were reported by Finney et al. (2016). The same biomass production pattern for foxtail millet monoculture vs. MIX3 and MIX6 was observed at the East 2017 site (Table 2), but not the same soil $\text{NO}_3\text{-N}$ differential (Table 6), perhaps because soil N levels overall were greater, or because the cover crop biomass was overall less compared to East 2016, leading to less competition.

Soil $\text{NO}_3\text{-N}$ was reduced by 17% in irrigated plots compared to non-irrigated plots at Fargo during 2017 (Table 6). Because differential soil moisture was not achieved due to limited supplemental irrigation at Absaraka in 2016, we would not expect to see irrigation treatment differences at this site. During 2017, reduction of soil nutrients associated with irrigation might be attributed to either greater uptake rates associated with increased nutrient movement in the soil, but also potentially increased cover crop root development due to increased water availability. Another factor affecting more mobile nutrients like N or K could be increased nutrient leaching because of increased soil moisture. For example,

Klocke et al. (1999) found that even when managed according to best management practices, irrigating corn (*Zea mays* L.) and soybean [*Glycine max* (L.) Merr.] in Nebraska resulted in soil N losses because of leaching.

5 | CONCLUSION

Integration of cover crops in cropping systems of the semi-arid and cold climate of the NGP is challenging due to the short growing season and erratic precipitation. In our study, multi-species cover crop mixtures did not produce more biomass than the most productive monocultures. Mixtures did, however, produce with less variability across differential water input (precipitation and irrigation) and climatic conditions. Therefore, multi-species cover crop mixtures, when grown as spring-planted, full-season service crops within a crop rotation, may offer multiple ecological and agronomic benefits while enhancing or stabilizing biomass production, thereby justifying the more complex management and additional expense associated with their use. Given the inconsistency in precipitation in the semi-arid NGP and the importance of realizing a return on investment when adopting the use of cover crops, planting multi-species mixtures may offer a more risk-averse approach when reliable productivity is the goal. We focused primarily on assessing biomass production and stability, but further studies should include evaluation of specific ecological and agronomic services to determine whether mixtures can provide these services at greater levels or with more consistency than single species plantings. Producers and other practitioners should also consider the specific services provided by individual cover crop species in relation to their goals, and cover crop selection should be site and region specific to maximize productivity for a given location when assembling cover crop mixtures. Finally, greater understanding of niche differentiation among cover crop species grown in mixtures should inform strategically designed cover crop mixtures capable of true transgressive overyielding (Wendling et al., 2017).

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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REFERENCES

- Allen, R. G., Walter, I. A., Elliot, R. L., & Howell, T. A. (2005). *The ASCE standardized reference evapotranspiration equation*. Reston, VA: ASCE Press.
- Ashton, I. W., Miller, A. E., Bowman, W. D., & Suding, K. N. (2010). Niche complementarity due to plasticity in resource use: Plant partitioning of chemical N forms. *Ecology*, 91, 3252–3260. <https://doi.org/10.1890/09-1849.1>
- Baltensperger, D. D. (1996). Foxtail and proso millet. In J. Janick (ed.) *Progress in new crops* (pp. 182–190). Alexandria, VA: ASHS Press.
- Baraibar, B., Hunter, M. C., Schipanski, M. E., Hamilton, A., & Mortensen, D. A. (2018). Weed suppression in cover crop monocultures and mixtures. *Weed Science*, 66, 121–133. <https://doi.org/10.1017/wsc.2017.59>
- Biederbeck, V. O., Bouman, O. T., Campbell, C. A., Winkleman, G. E., & Bailey, L. D. (1996). Nitrogen benefits from four green-manure legumes in dryland cropping systems. *Canadian Journal of Plant Science*, 76, 307–315. <https://doi.org/10.4141/cjps96-053>
- Blanco-Canqui, H., Holman, J. D., Schlegel, A. J., Tatarko, J., & Shaver, T. M. (2013). Replacing fallow with cover crops in a semiarid soil: Effects on soil properties. *Soil Science Society of America Journal*, 77, 1026–1034. <https://doi.org/10.2136/sssaj2013.01.0006>
- Carr, P., Anderson, R., Lawley, Y., Miller, P., & Zwinger, S. (2012). Organic zero-till in the northern US Great Plains Region: Opportunities and obstacles. *Renewable Agriculture & Food Systems*, 27, 12–20. <https://doi.org/10.1017/S174217051100041X>
- Chakwizira, E., Johnstone, P., Fletcher, A. L., Meenken, E. D., de Ruiter, J. M., & Brown, H. E. (2014). Effects of nitrogen rate on nitrate-nitrogen accumulation in forage kale and rape crops. *Grass & Forage Science*, 70, 268–282. <https://doi.org/10.1111/gfs.12109>
- Clark, A. (2007). *Managing cover crops profitably* (3rd ed., Handbook Series Book 9). College Park, MD: Sustainable Agriculture Research and Education, University of Maryland.
- Conservation Technology Information Center (CTIC) (2017). *Report of the 2016-17 National Cover Crop Survey*. West Lafayette, IN: Conservation Technology Information Center, the North Central Region Sustainable Agriculture Research and Education Program, and the American Seed Trade Association. Conservation Technology Information Center. <https://www.sare.org/wp-content/uploads/2016-2017-Cover-Crop-Survey-Report.pdf>
- Conservation Technology Information Center (CTIC) (2020). *Report of the 2019-20 National Cover Crop Survey*. West Lafayette, IN: Conservation Technology Information Center, the North Central Region Sustainable Agriculture Research and Education Program, and the American Seed Trade Association. Conservation Technology Information Center. <https://www.ctic.org/files/CoverCropSurvey%202020%2024mb.pdf>
- Creamer, N. G., Bennett, M. A., & Stinner, B. R. (1997). Evaluation of cover crop mixtures for using in vegetable production systems. *Hortscience*, 32, 866–870. <https://doi.org/10.21273/HORTSCI.32.5.866>
- Dabney, S. M., Delgado, J. A., & Reeves, D. W. (2001). Using winter cover crops to improve soil and water quality. *Communications in Soil Science & Plant Analysis*, 32, 1221–1250. <https://doi.org/10.1081/CSS-100104110>
- Diaz, S., & Cabido, M. (2001). Vive la difference: Plant functional diversity matters to ecosystem processes. *Trends in Ecology & Evolution*, 16, 646–655. [https://doi.org/10.1016/S0169-5347\(01\)02283-2](https://doi.org/10.1016/S0169-5347(01)02283-2)
- Finney, D. M., & Kaye, J. P. (2017). Functional diversity in cover crop polycultures increases multifunctionality of an agricultural system. *Journal of Applied Ecology*, 54, 509–517. <https://doi.org/10.1111/1365-2664.12765>
- Finney, D. M., Murrell, E. G., White, C. M., Baraibar, B., Barbercheck, M. E., Bradley, B. A., Cornelisse, S., Hunter, M. C., Kaye, J. P., Mortensen, D. A., Mullen, C. A., & Schipanski, M. E. (2017). Ecosystem services and disservices are bundled in simple and diverse cover cropping systems. *Agricultural & Environmental Letters*, 2, 170033. <https://doi.org/10.2134/ael2017.09.0033>
- Finney, D. M., White, C. M., & Kaye, J. P. (2016). Biomass production and carbon/nitrogen ratio influence ecosystem services from cover crop mixtures. *Agronomy Journal*, 108, 39–52. <https://doi.org/10.2134/agronj15.0182>
- Fisk, J. W., Hesterman, O. B., Shrestha, A., Kells, J. J., Harwood, R. R., Squire, J. M., & Sheaffer, C. C. (2001). Weed suppression by annual legume over crops in no-tillage corn. *Agronomy Journal*, 93, 319–325. <https://doi.org/10.2134/agronj2001.932319x>
- Florence, A. M., Higley, L. G., Drijber, R. A., Francis, C. A., & Lindquist, J. L. (2019). Cover crop mixture diversity, biomass productivity, weed suppression, and stability. *Plos One*, 14(3), e0206195. <https://doi.org/10.1371/journal.pone.0206195>
- Franco, J. G., King, S. R., & Volder, A. (2018). Component crop physiology and water use efficiency in response to intercropping. *European Journal of Agronomy*, 93, 27–39. <https://doi.org/10.1016/j.eja.2017.11.005>
- Frank, D. A., & McNaughton, S. J. (1991). Stability increases with diversity in plant communities: Empirical evidence from the 1988 Yellowstone drought. *Oikos*, 62, 360–362. <https://www.jstor.org/stable/3545501>
- Franzen, D. (2016). Fertilizing sunflower (Publication SF713). North Dakota State University Extension. Fargo, ND. <https://www.ag.ndsu.edu/publications/crops/fertilizing-sunflower/sf713.pdf> (accessed 6 Dec. 2020).
- Giller, K. E., Ormesher, J., & Awah, F. M. (1991). Nitrogen transfer from *Phaseolus* bean to intercropped maize measured using ¹⁵N-enrichment and ¹⁵N-isotope dilution methods. *Soil Biology & Biochemistry*, 23, 339–346. [https://doi.org/10.1016/0038-0717\(91\)90189-Q](https://doi.org/10.1016/0038-0717(91)90189-Q)

- Haramoto, E. R., & Gallandt, E. R. (2004). Brassica cover cropping for weed management: A review. *Renewable Agriculture & Food Systems*, 19, 187–198. <https://doi.org/10.1079/RAFS200490>
- Haramoto, E. R., & Gallandt, E. R. (2005). Brassica cover cropping: I. Effects on weed and crop establishment. *Weed Science*, 53, 695–701. <https://doi.org/10.1614/WS-04-162R.1>
- Hendrickson, J. R., Schmer, M. R., & Sanderson, M. A. (2013). Water use efficiency by switchgrass compared to a native grass or a native grass alfalfa mixture. *BioEnergy Research*, 6, 746–754. <https://doi.org/10.1007/s12155-012-9290-3>
- Hooper, D. U. (1997). The effects of plant composition and diversity on ecosystem processes. *Science*, 277, 1302–1305. <https://doi.org/10.1126/science.277.5330.1302>
- Hunter, M. C., Schipanski, M. E., Burgess, M. H., LaChance, J. C., Bradley, B. A., Barbercheck, M. E., Kaye, J. P., & Mortensen, D. A. (2019). Cover crop mixture effects on maize, soybean, and wheat yield in rotation. *Agricultural & Environmental Letters*, 4, 180051. <https://doi.org/10.2134/ael2018.10.0051>
- Janosevic, B., Dolijanovic, Z., Dragicevic, V., Simic, M., Dodevska, M., Djordjevic, S., Moravcevic, D.J., & Miodragovic, R. (2017). Cover crop effects on the fate of N in sweet maize (*Zea mays* L. *saccharate* Sturt.) production in a semiarid region. *International Journal of Plant Production*, 11, 1735–6814. <https://doi.org/10.22069/IJPP.2017.3425>
- Khan, Q. A., & Mcvay, K. A. (2019). Productivity and stability of multi-species cover crop mixtures in the Northern Great Plains. *Agronomy Journal*, 111, 1817–1827. <https://doi.org/10.2134/agronj2018.03.0173>
- Klocke, N. L., Watts, D. G., Schneekloth, J. P., Davison, D. R., Todd, R. W., & Parkhurst, A. M. (1999). Nitrate leaching in irrigated corn and soybean in a semi-arid climate. *Transactions of the ASAE*, 42, 1621–1630. <https://doi.org/10.13031/2013.13328>
- Liebig, M. A., Hendrickson, J. R., Archer, D. W., Schmer, M. A., Nichols, K. A., & Tanaka, D. L. (2015). Short-term soil responses to late-seeded cover crops in a semi-arid environment. *Agronomy Journal*, 107, 2011–2019. <https://doi.org/10.2134/agronj15.0146>
- Liu, A., Ma, B. L., & Bomke, A. A. (2005). Effects of cover crops on soil aggregate stability, total organic carbon, and polysaccharides. *Soil Science Society of America Journal*, 69, 2041–2048. <https://doi.org/10.2136/sssaj2005.0032>
- Mallinger, R. E., Franco, J. G., Prischmann-Voldseth, D. A., & Prasifka, J. R. (2019). Annual cover crops for managed and wild bees: Optimal plant mixtures depend on pollinator enhancement goals. *Agriculture, Ecosystems & Environment*, 273, 107–116. <https://doi.org/10.1016/j.agee.2018.12.006>
- Matsuura, A., & An, P. (2020). Factors related to water and dry matter during pre- and post-heading in four millet species under server water deficit. *Plant Production Science*, 23, 28–38. <https://doi.org/10.1080/1343943X.2019.1698969>
- Mazzoncini, M., Sapkota, T. B., Barberi, P., Antichi, D., & Risaliti, R. (2011). Long-term effect of tillage, nitrogen fertilization and cover crops on soil organic carbon and total nitrogen content. *Soil & Tillage Research*, 114, 165–174. <https://doi.org/10.1016/j.still.2011.05.001>
- Mccann, K. S. (2000). The diversity-stability debate. *Nature*, 405, 228–233. <https://doi.org/10.1038/35012234>
- Mitchell, J. P., Shrestha, A., & Irmak, S. (2015). Trade-offs between winter cover crop production and soil water depletion in the San Joaquin Valley, California. *Journal of Soil & Water Conservation*, 70, 430–440. <https://doi.org/10.2489/jswc.70.6.430>
- Monfort, W. S., Csinos, A. S., Desaegeer, J., Seebold, K., Webster, T. M., & Diaz-Perez, J. C. (2007). Evaluating *Brassica* species as an alternative control measure for root-knot nematode (*M. incognita*) in Georgia vegetable plasticulture. *Crop Protection*, 26, 1359–1368. <https://doi.org/10.1016/j.cropro.2006.11.008>
- Moroke, T. S., Schwartz, R. C., Brown, K. W., & Juo, A. S. R. (2011). Water use efficiency of dryland cowpea, sorghum and sunflower under reduced tillage. *Soil & Tillage Research*, 112, 76–84. <https://doi.org/10.1016/j.still.2010.11.008>
- Nadeem, F., Ahmad, Z., Ul Hassan, M., Wang, R., Diao, X., & Li, X. (2020). Adaptation of foxtail millet (*Setaria italica* L.) to abiotic stresses: A special perspective of responses to nitrogen and phosphate limitation. *Frontiers in Plant Science*, 11, 187. <https://doi.org/10.3389/fpls.2020.00187>
- Nematpour, A., Eshghizadeh, H. R., & Zahedi, M. (2019). Drought-tolerance mechanisms in foxtail millet (*Setaria italica*) and proso millet (*Panicum miliaceum*) under different nitrogen supply and sowing dates. *Crop & Pasture Science*, 70, 442–452. <https://doi.org/10.1071/CP18501>
- Nielsen, D. C., Lyon, D. J., Hergert, G. W., Higgins, R. K., Calderón, F. J., & Vigil, M. F. (2015b). Cover crop mixtures do not use water differently than single-species plantings. *Agronomy Journal*, 107, 1025–1038. <https://doi.org/10.2134/agronj14.0504>
- Nielsen, D. C., Lyon, D. J., Hergert, G. W., Higgins, R. K., & Holman, J. D. (2015a). Cover crop biomass production and water use in the central Great Plains. *Agronomy Journal*, 107, 2047–2058. <https://doi.org/10.2134/agronj15.0186>
- Nielsen, D. C., Lyon, D. J., Higgins, R. K., Hergert, G. W., Holman, J. D., & Vigil, M. F. (2016). Cover crop effect on subsequent wheat yield in the Central Great Plains. *Agronomy Journal*, 108, 243–256. <https://doi.org/10.2134/agronj2015.0372>
- Nielsen, D. C., & Vigil, M. F. (2005). Legume green fallow effect on soil water content at wheat planting and wheat yield. *Agronomy Journal*, 97, 684–689. <https://doi.org/10.2134/agronj2004.0071>
- Norsworthy, J. K., McClelland, M., Griffith, G., Bangarwa, S. K., & Still, J. (2011). Evaluation of cereal and Brassicaceae cover crops in conservation-tillage, enhanced, glyphosate-resistant cotton. *Weed Technology*, 25, 6–13. <https://doi.org/10.1614/WT-D-10-00040.1>
- Ofori, C. F., & Stern, W. R. (1987). Cereal-legume intercropping systems. *Advances in Agronomy*, 26, 177–204. [https://doi.org/10.1016/S0065-2113\(08\)60802-0](https://doi.org/10.1016/S0065-2113(08)60802-0)
- Padbury, G., Waltman, S., Caprio, J., Coen, G., McGinn, S., Mortensen, D., Nielsen, G., & Sinclair, R. (2002). Agroecosystems and land resources of the northern Great Plains. *Agronomy Journal*, 94, 251–261. <https://doi.org/10.2134/agronj2002.2510>
- Sanderson, M., Johnson, H., & Hendrickson, J. (2018). Cover crop mixtures grown for annual forage in a semi-arid environment. *Agronomy Journal*, 110, 525–534. <https://doi.org/10.2134/agronj2017.04.0228>
- Schipanski, M. E., Barbercheck, M., Douglas, M. R., Finney, D. M., Haider, K., Kaye, J. P., Kemanian, A. R., Mortensen, D. A., Ryan, M. R., Tooker, J., & White, C. (2014). A framework for evaluating ecosystem services provided by cover crops in agroecosystems. *Agricultural Systems*, 125, 12–22. <https://doi.org/10.1016/j.agsy.2013.11.004>
- Schläpfer, F., & Schmid, B. (1999). Ecosystem effects on biodiversity: A classification of hypotheses and exploration of empirical results. *Ecological Applications*, 9, 893–912. [https://doi.org/10.1890/1051-0761\(1999\)009\[0893:EEOBAC\]2.0.CO;2](https://doi.org/10.1890/1051-0761(1999)009[0893:EEOBAC]2.0.CO;2)

- Schmid, B., Hector, A., Saha, P., & Loreau, M. (2008). Biodiversity effects and transgressive overyielding. *Journal of Plant Ecology*, *1*, 95–102. <https://doi.org/10.1093/jpe/rtn011>
- Seigies, A. T., & Pritts, M. (2006). Cover crop rotations alter soil microbiology and reduce replant disorders in strawberry. *Hortscience*, *41*, 1303–1308. <https://doi.org/10.21273/HORTSCI.41.5.1303>
- Sinclair, T. R., Tanner, C. B., & Bennett, J. M. (1984). Water-use efficiency in crop production. *Bioscience*, *34*, 36–40. <https://doi.org/10.2307/1309424>
- Smith, R. G., Atwood, L. W., & Warren, N. D. (2014). Increased productivity of a cover crop mixture is not associated with enhanced agroecosystem services. *PLOS ONE*, *9*, e97351. <https://doi.org/10.1371/journal.pone.0097351>
- Snapp, S. S., Swinton, S. M., Labarta, R., Mutch, D., Black, J. R., Leep, R., Nyiraneza, J., & O'Neil, K. (2005). Evaluating cover crops for benefits, costs and performance within cropping systems niches. *Agronomy Journal*, *97*, 322–332. <https://doi.org/10.2134/agronj2005.0322a>
- St Aime, R., Zehnder, G. W., Talley, C., & Narayanan, S. (2020). Differences in biomass production and water use efficiency among seven different cover crops in the wet winter seasons of 2016/17 and 2018 in South Carolina. *Agronomy*, *10*, 463. <https://doi.org/10.3390/agronomy10040463>
- Teasdale, J. R. (1996). Contribution of cover crops to weed management in sustainable agricultural systems. *Journal of Production Agriculture*, *9*, 475–479. <https://doi.org/10.2134/jpa1996.0475>
- Tilman, D. (1996). Biodiversity: Population versus ecosystem stability. *Ecology*, *77*, 350–363. <https://doi.org/10.2307/2265614>
- Tilman, D. (1997). The influence of functional diversity and composition on ecosystem processes. *Science*, *277*, 1300–1302. <https://doi.org/10.1126/science.277.5330.1300>
- Tilman, D., & Downing, J. A. (1994). Biodiversity and stability in grasslands. *Nature*, *367*, 363–365. <https://doi.org/10.1038/367363a0>
- Tonitto, C., David, M. B., & Drinkwater, L. E. (2006). Replacing bare fallows with cover crops in fertilizer-intensive cropping systems: A meta-analysis of crop yield and N dynamics. *Agriculture, Ecosystems & Environment*, *112*, 58–72. <https://doi.org/10.1016/j.agee.2005.07.003>
- Tyson, S. C., & Cabrera, M. L. (1993). Nitrogen mineralization in soils amended with composted and uncomposted poultry litter. *Communications in Soil Science & Plant Analysis*, *24*, 2361–2374. <https://doi.org/10.1080/0010362930936896>
- Unger, P. W., Fryrear, D. W., & Lindstrom, M. J. (2006). Soil conservation. In G. A. Peterson, P. W. Unger, & W. A. Payne, (Eds.), *Dryland agriculture* (pp. 87–112). Madison, WI: ASA, CSSA, and SSSA.
- Unger, P. W., & Vigil, M. F. (1998). Cover crop effects on soil water relationships. *Journal of Soil & Water Conservation*, *53*, 200–207.
- Wendling, M., Buchi, L., Amosse, C., Jeangros, B., Walter, A., & Charles, R. (2017). Specific interactions leading to transgressive overyielding in cover crop mixtures. *Agriculture, Ecosystems & Environment*, (241), 88–99. <https://doi.org/10.1016/j.agee.2017.03.003>
- Wendling, M., Buchi, L., Amosse, C., Sinaj, S., Walter, A., & Charles, R. (2016). Influence of root and leaf traits on the uptake of nutrients in cover crops. *Plant & Soil*, *409*, 419–434. <https://doi.org/10.1007/s11104-016-2974-2>
- Wiggins, M. S., McClure, M. A., Hayes, R. M., & Steckel, L. E. (2015). Integrating cover crops and POST herbicides for glyphosate-resistant palmer amaranth (*Amaranthus palmeri*) control in corn. *Weed Technology*, *29*, 412–418. <https://doi.org/10.1614/WT-D-14-00145.1>
- Wortman, S. E., Francis, C. A., & Lindquist, J. L. (2012). Cover crop mixtures for the Western Corn Belt: Opportunities for increased productivity and stability. *Agronomy Journal*, *104*, 699–705. <https://doi.org/10.2134/agronj2011.0422>

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